机电系统建模、辨识与控制
Modeling, Identification and Control of Mechatronic Systems

MIC_L4 System Identification—— Principle

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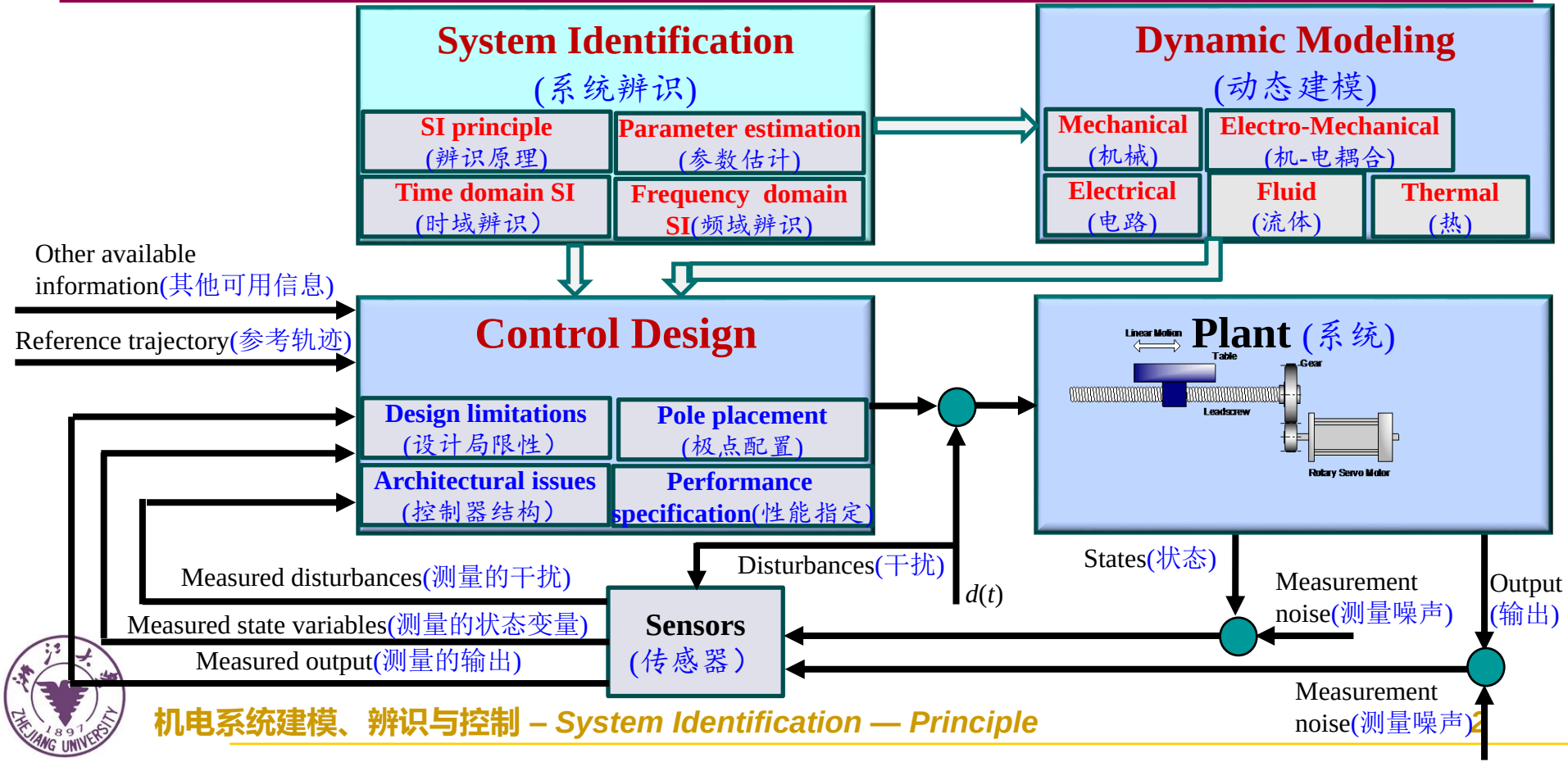
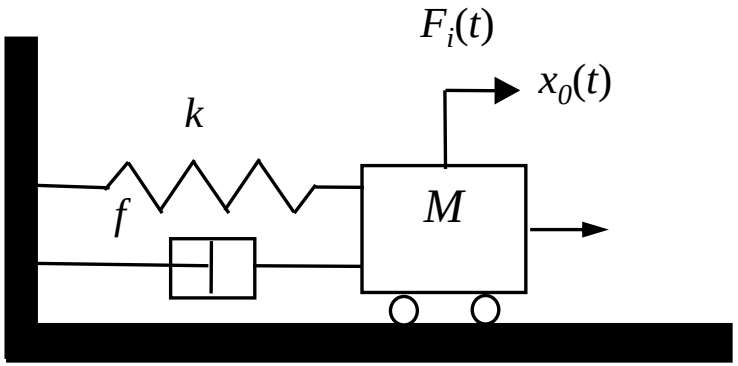
Main Contents of Course(课程内容)

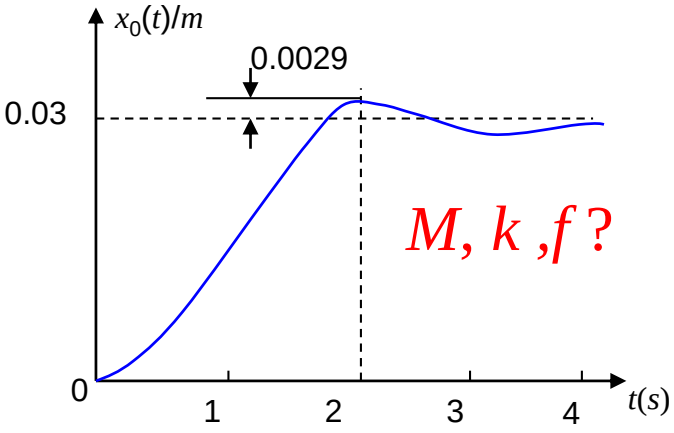


Examples of System Identification

A mass-spring-damping system with applied step force 8.9N and recording its step response. Please calculate M、k and damping coefficient f. (弹簧-质量-阻尼系统通过阶跃响应的辨识)

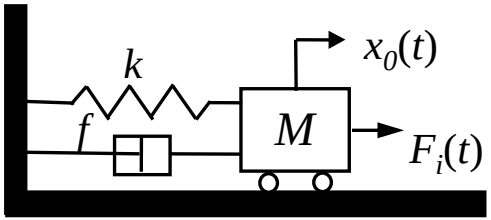


Schematics

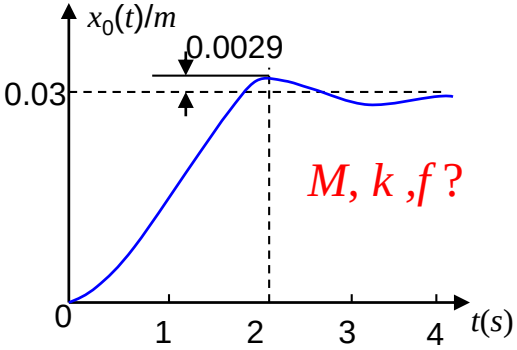


Step response

Examples of System Identification



Schematics



Step response

System dynamics $$F_i(t) - kx_0(t) - f\frac{dx_0(t)}{dt} = M\frac{dx_0^2(t)}{dt^2}$$

System output $$X_0(s) = \frac{1}{Ms^2 + fs + k}\cdot F_i(s) = \frac{1}{Ms^2 + fs + k}\cdot\frac{8.9}{s}$$

Steady state value $$\lim_{t\to\infty} x_0(t) = \lim_{s\to 0} sX_0(s) = \frac{8.9}{k} = 0.03(m)$$

$$\Rightarrow k = 297(N/m)$$

Overshoot $$M_p = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} = \frac{0.0029}{0.03}$$

$$\Rightarrow \zeta = 0.6$$

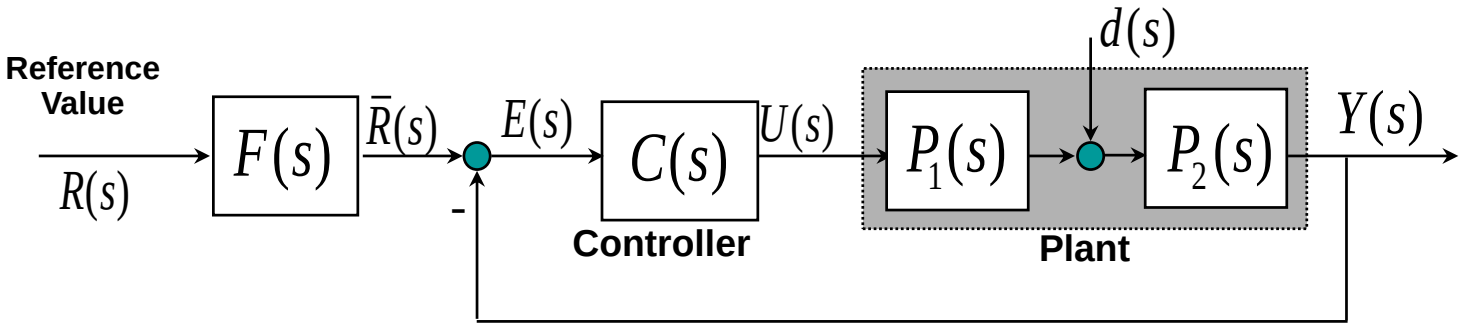
Natural frequency $$\omega_n = \frac{\pi}{t_p\sqrt{1-\zeta^2}} = \frac{\pi}{2\sqrt{1-0.6^2}} = 1.96(rad/s)$$

$$\Rightarrow M = \frac{k}{\omega_n^2} = 77.3(kg)$$

$$f = 2\zeta\omega_n M = 181.8(N\cdot m/s)$$

Obtaining the model parameters from step response via open control!（通过开环控制的阶跃响应获得模型参数）

Concept of System Identification



- **Dynamic System**

$$y = \mathrm{H}(u, w), u \in \Omega_u, w \in \Omega_w, \mathrm{H} : \ell_{\infty e} \times \ell_{\infty e} \to \ell_{\infty e}$$

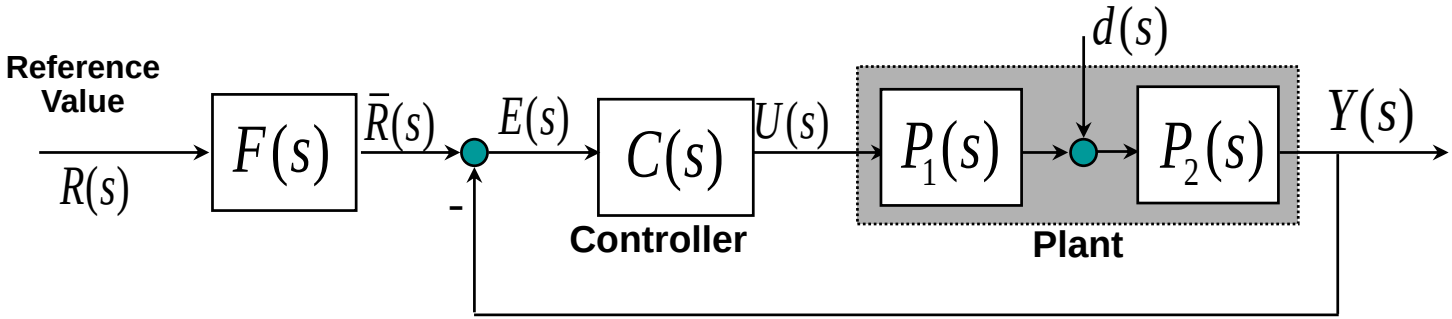
y: output

u: input and/or measured disturbance

w: unmeasured disturbance

- **System Identification**
  - ➢ Obtaining the mapping *H* based on experimental data collected through conducting certain experiments (基于一定实验收集的实验数据来获得映射)
    - – First step in any model-based control design process (基于模型设计过程的第一步)
    - – Quality of the resulting model has significant practical importance to the achievable closed-loop control performance (所获得模型品质对可达到的闭环控制性能具有非常重要的作用)

Concept of System Identification



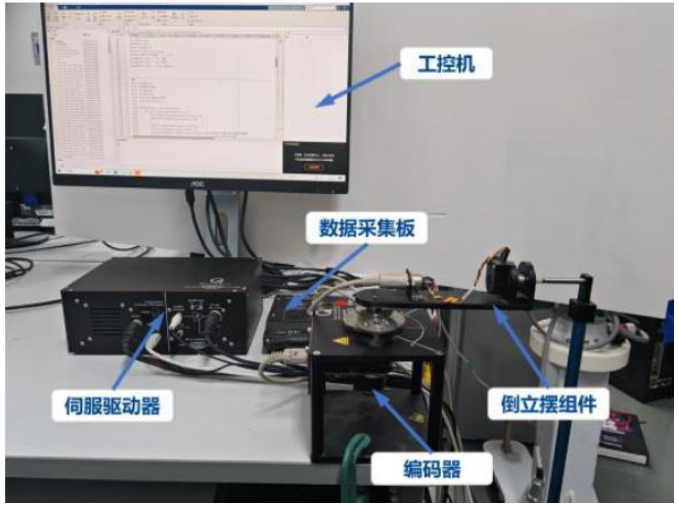
- **Dynamic System**

$$y = \mathrm{H}(u, w), u \in \Omega_u, w \in \Omega_w, \mathrm{H} : \ell_{\infty e} \times \ell_{\infty e} \to \ell_{\infty e}$$

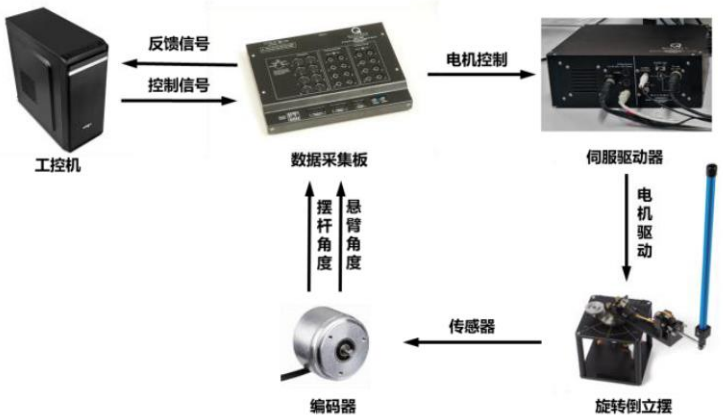
- **Implementation**
  - Constructing an estimation of H based on the collected experimental data set $Z_t = \{U_t, Y_t\}$ at sampling instant time t, in which $U_t = \{u(t), u(t-1), \ldots\}$ and $Y_t = \{y(t), y(t-1), \ldots\}$ (基于采样的实验数据计算H 的估计量)

Schematics of Rotary Inverted Pendulum

- 倒立摆系统组成
  - 倒立摆
  - 伺服电机
  - 齿轮减速器
  - 电流驱动器
  - CRIO控制器
  - 编码器



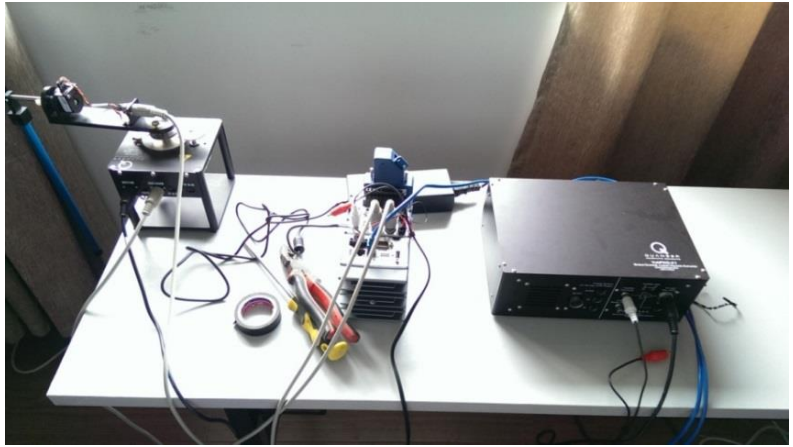
旋转倒立摆实验系统



旋转倒立摆工作原理示意图

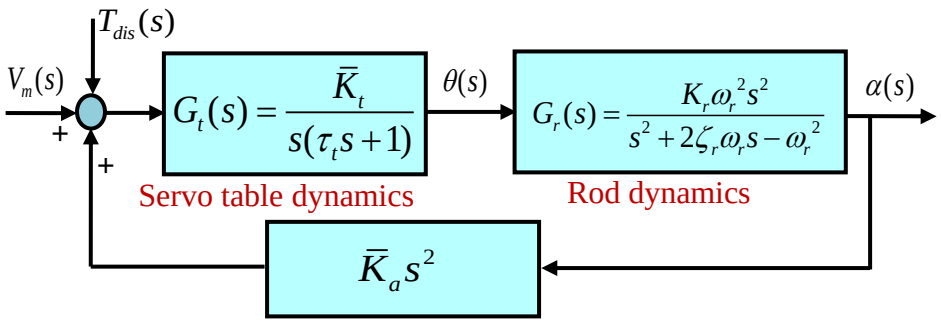
System Identification in Time Domain

- **System Identification via Open-Loop Experiments**（通过开环实验进行系统辨识）
  - Open-loop experiments can only be performed for **stable** systems only（开环实验仅能适用稳定系统）
  - Inverted pendulum system is inherently **unstable**（倒立摆系统本质上不稳定）
  - Only use open-loop experiments to identify **stable subsystem** dynamics of the servo table without rod（仅采用开环实验来辨识不带摆杆的伺服转台的稳定子动力学）

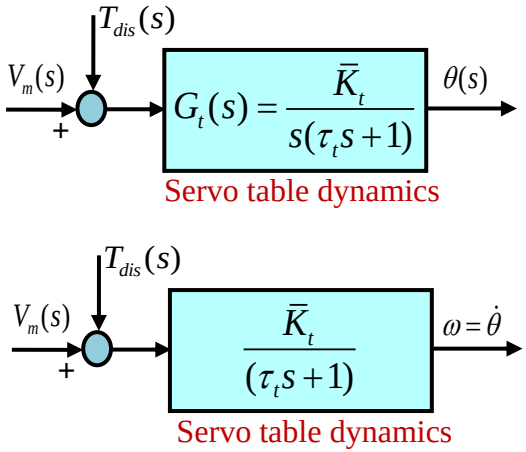


System Identification of Rotary Servo Table

- **Servo Table without Rod**(不带摆杆的伺服转台)



Block diagram of the Rotary inverted pendulum system



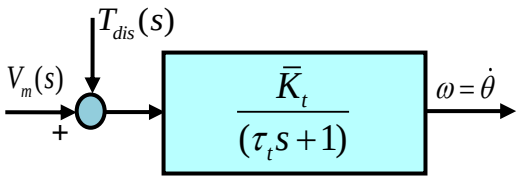
$$\zeta_r=\sqrt{\frac{B_p^2}{4I_p m_p g L_c}},\ \omega_r=\sqrt{\frac{m_p g L_c}{I_p}},\ K_r=\frac{L_t}{g},\ \tau_t=\frac{I_t}{B_t},\ \bar{K}_t=\frac{K_u}{B_t},\ \bar{K}_a=\frac{m_p L_t L_c}{K_u}$$

$$I_t=J_t+m_p L_t^2,\ I_p=J_p+m_p L_c^2,\ K_u=\frac{\eta_m\eta_g k_t K_g}{R_m}$$

Parameters of Nominal System

➢ True value of the nominal system

Symbol	Description	Value	Unit	Variance
L_t	Rotary arm length from pivot to tip	0.216	m	
J_r	Rotary arm moment of inertia about its center of mass	0.001	kg.m^2	
B_r	Rotary arm viscous damping coefficient as seen at the pivot axis	0.0024	N.m.s/rad	
R_m	Motor armature resistance	2.6	Ω	12%
k_t	Motor current-torque constant	7.68×10^{-3}	N.m/A	12%
k_m	Motor back-EMF constant	7.68×10^{-3}	V/(rad/s)	12%
K_g	Total gear ratio	14		
η_m	Motor efficiency	0.69		5%
η_g	Gearbox efficiency	0.9		10%
V_m	Applied voltage to the motor	-10~10	V	
A_c	Coulomb friction magnitude	0.9		$T_{dis}=A_c\mathrm{sign}(\omega)$



Servo table dynamics

Model parameters of servo table without rod

$$\bar{K}_t = 4.7688$$
$$\tau_t = 0.1920$$
$$A_c = 0.9$$

System ID of Servo Table Without Rod

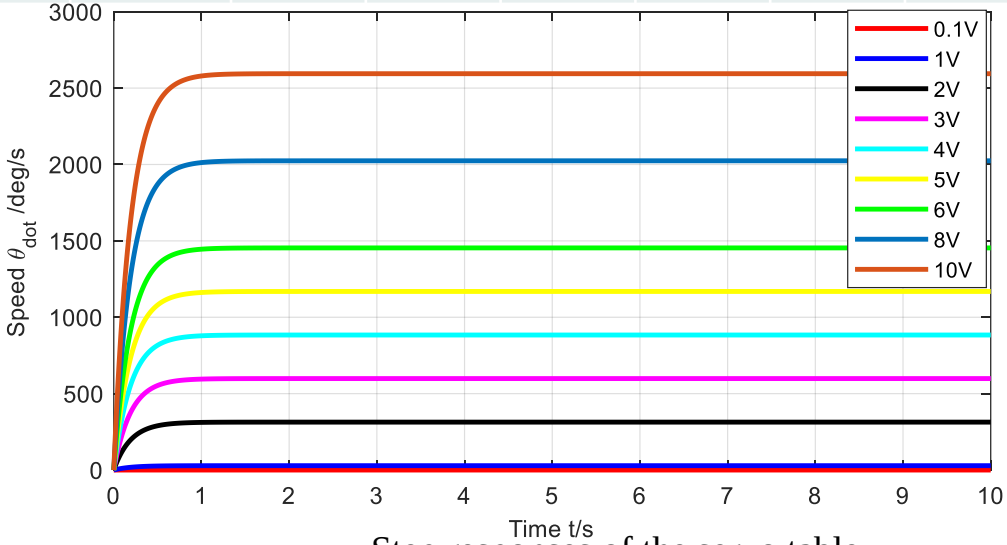
- **Method 1: System Identification in Time-Domain via Step Response**
  - **Identification via Step Responses without Considering Coulomb Friction**

u_c	TRUE	0.1	1	2	3	4	5	6	8	10
K_t	4.9757	0.0011	0.4345	2.6751	3.4417	3.8253	4.0553	4.2087	4.4006	4.5156
τ_t	0.1938	0.002	0.156	0.188	0.190	0.192	0.192	0.194	0.194	0.194

Mean Value

$\bar{K}_t = 3.8747$

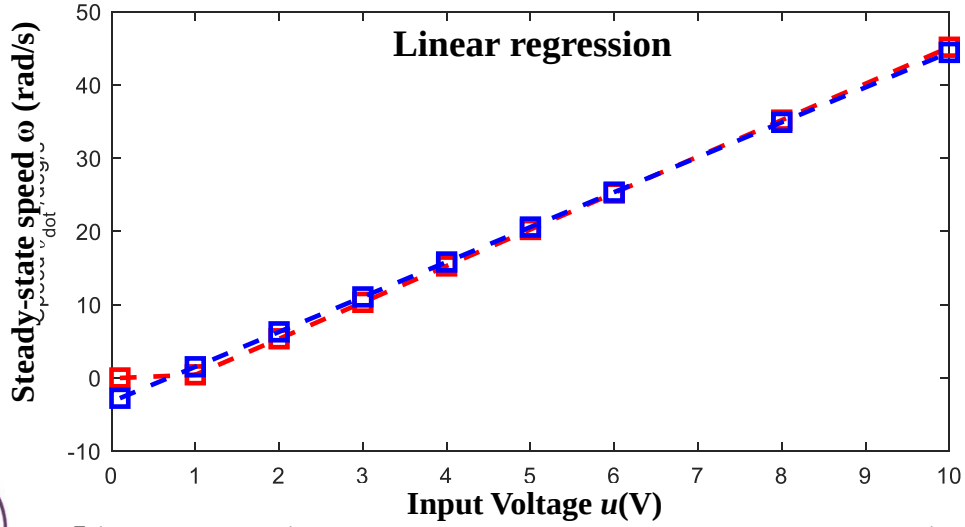
$\tau_t = 0.1920$



Step responses of the servo table

System ID of Servo Table Without Rod

- **Identification via Step Responses with Considering Coulomb Friction Model**
  - Stiction level can be obtained via simple experiments of gradually increasing the input voltage until table movement（逐渐增加输入电压直至伺服转台运动, 从而获得静摩擦力）
  - Model parameters K_t and τ_t can then be obtained via experimental step responses assuming Coulomb friction the same as the identified stiction（假设库伦摩擦力为辨识出的静摩擦力，并通过阶跃响应实验获得模型参数K_t和τ_t）



Linear regression of steady-state response values vs step input voltages

$$\omega_{ss} = 4.7688(u - 0.6794)$$

Estimated Value

$\bar{K}_t = 4.7688$

$T_{dis} = 0.6794\,\mathrm{sgn}(\omega)$

$\tau_t = 0.1920$

True Value

$\bar{K}_{t0} = 4.9757$

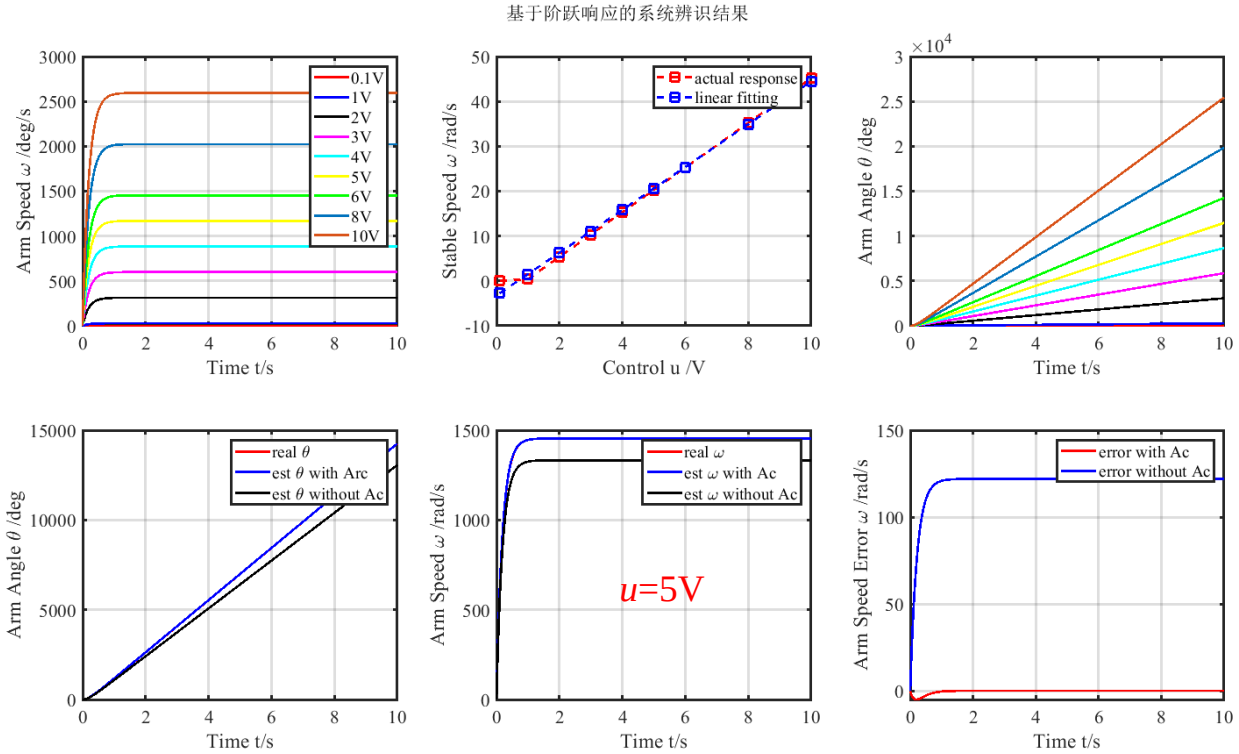
$\tau_{t0} = 0.1938$

$T_{dis0} = 0.9\,\mathrm{sgn}(\omega)$

System ID of Servo Table Without Rod

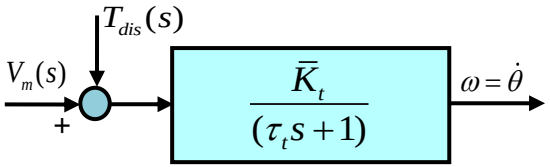
- **Identification result via Step Responses**

u_c	K_t	τ_t	A_c
True value	4.9757	0.1938	0.9
0.1	0.0011	0.002	
1	0.4345	0.156	
2	2.6751	0.188	
3	3.4417	0.190	
4	3.8253	0.192	
5	4.0553	0.192	
6	4.2087	0.194	
8	4.4006	0.194	
10	4.5156	0.194	
With A_c	3.8747	0.1920	0
Without A_c	4.7688	0.1920	0.6794



System ID of Servo Table Without Rod

- **Method 2: System Identification in Time-Domain via LSE**
  - **Deficiency of System Identification via Step response?**
    - Use specific model structure which can be simplified as one-order or two-order system (能简化为一阶或二阶系统的特定模型结构)
    - Need stable response (需要稳定响应)
    - No systematic procedure: multiple steps for system identification (需要多步骤实现系统辨识)



Servo table dynamics

  - **Simultaneously obtain all model parameters via the least squares estimation (LSE) algorithm** (通过最小二乘方法同时获得所有的模型参数)

System Identification in Time-Domain via LSE

■ ID Step 1 — Model Structure Selection (模型结构选择)

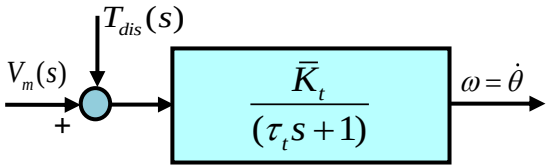
- **Use parameterized model for Coulomb friction, e.g.** , (采用库伦摩擦力的参数化模型)

$$\bar{T}_c = A_c \mathrm{sign}(\omega)$$

- **Obtain parameterized dynamic model for system ID** (获得用于系统辨识的参数化动态模型)

$$\tau_t \dot{\omega} + \omega = K_t [u - A_c sign(\omega)]$$

$$\Rightarrow \quad \underbrace{\frac{\tau_t}{\bar{K}_t}}_{\vartheta_1} \dot{\omega} + \underbrace{\frac{1}{\bar{K}_t}}_{\vartheta_2} \omega + \underbrace{A_c}_{\vartheta_3} \mathrm{sign}(\omega) = u \quad \Leftrightarrow \quad \underbrace{\begin{bmatrix} \dot{\omega} & \omega & \mathrm{sign}(\omega) \end{bmatrix}}_{\phi(t)^T} \underbrace{\begin{bmatrix} \vartheta_1 \\ \vartheta_2 \\ \vartheta_3 \end{bmatrix}}_{\vartheta} = \underbrace{u}_{y} \quad \Rightarrow \quad \boxed{\phi(t)^T \vartheta = y}$$



Servo table dynamics

Note: Original dynamic model could be nonlinear as long as the regressor can be calculated based on experimentally obtained data! (只要基于实验数据能计算出回归量，原始的动态模型可以是非线性的)

System ID of Servo Table Without Rod

■ ID Step 2 — Criterion of Fit（拟合准则）

➢ **Least Square Estimation (LSE) is to minimize the sum of the squares of the model prediction errors over the entire experimental time {*t*=*kT*, *k*=1,…,*N*}**（LSE用来最小化整个实验阶段的模型预测误差的和）

- **Prediction error at each sampling instance *t*=*kT***（每个采样时刻的预测误差）

$$\varepsilon_\vartheta(t) = \underbrace{y(t)}_{\text{actual measured value}} - \underbrace{\phi(t)^T \hat{\vartheta}}_{\text{predicted value with model parameters } \vartheta}$$

- **Weighted Fitting-Error-Squares Type Performance Index Function**（加权拟合误差平方型性能指标函数）

$$J(t,\vartheta) = \sum_{t=1}^{N} [R(t)\varepsilon_\vartheta(t)]^2 = \sum_{t=1}^{N} \varepsilon_\vartheta(t)^T \underbrace{R(t)^T R(t)}_{W(t)} \varepsilon_\vartheta(t)$$

$$\Phi = \begin{bmatrix} \phi(1) \\ \dots \\ \phi(k) \\ \dots \\ \phi(N) \end{bmatrix} \in R^{N\times 3}, Y = \begin{bmatrix} y(1) \\ \dots \\ y(k) \\ \dots \\ y(N) \end{bmatrix} \in R^N$$

$$J(t,\theta) = [Y(t) - \Phi(t)\theta]^T W(t)[Y(t) - \Phi(t)\theta]$$

System ID of Servo Table Without Rod

■ ID Step 2 — Criterion of Fit (拟合准则)

$$J(t,\boldsymbol{\theta}) = [Y(t) - \Phi(t)\theta]^T W(t)[Y(t) - \Phi(t)\theta]$$

LSE algorithm

$$\left.\frac{\partial J(k,\vartheta)}{\partial \vartheta}\right|_{\vartheta=\hat{\vartheta}(t)} = 0$$

$$\hat{\vartheta}_{LSE}(t) = \underline{[\Phi^T(t)W(t)\Phi(t)]^{-1}}\Phi^T(t)W(t)Y(t) = P(t)\Phi^T(t)W(t)Y(t)$$

$\Phi(t)^T W(t)\, \Phi(t)$ is nonsingular, i.e., $\Phi(t)$ has a full column rank *p*

The solution of LSE exists only if :

E1. The data set has to be large enough

E2. The data set has to be rich enough

System ID of Servo Table Without Rod

■ ID Step 3 — Experimental Design

- **Actual Process** $y(t) = \varphi(t)^T \theta^0 + e(t)$ **e(t): model error**
- **LSE Estimation Error under Model Structure Error**

where, $R_{\varphi e}(\tau) = \bar{E}[\varphi(t)e(t-\tau)^T],\ \forall \tau$

$$\tilde{\theta}(t) = \hat{\theta}(t) - \theta^o = \left(\frac{1}{t}\sum_{i=1}^{t}\varphi(i)\varphi(i)^T\right)^{-1}\left(\frac{1}{t}\sum_{i=1}^{t}\varphi(i)e(i)\right) \xrightarrow{t\to\infty} R_\varphi(0)^{-1}R_{\varphi e}(0)$$

$$\hat{\theta}(t) = P(t)\Phi(t)^T Y(t) = P(t)[\sum_{i=1}^{t}\varphi(i)y(i)]$$

Uncorrelated $\varphi(i)$ and $e(i)$ with $E[e(t)] = 0$ → $R_{\varphi e}(0) = 0$ → An unbiased estimate (i.e., $\theta(t) \to \theta^0$ as $t \to \infty$) is obtained

Persistent exciting (PE) condition → $R_\varphi(0)$ is nonsingular → $R_{\varphi e}(0) = 0$

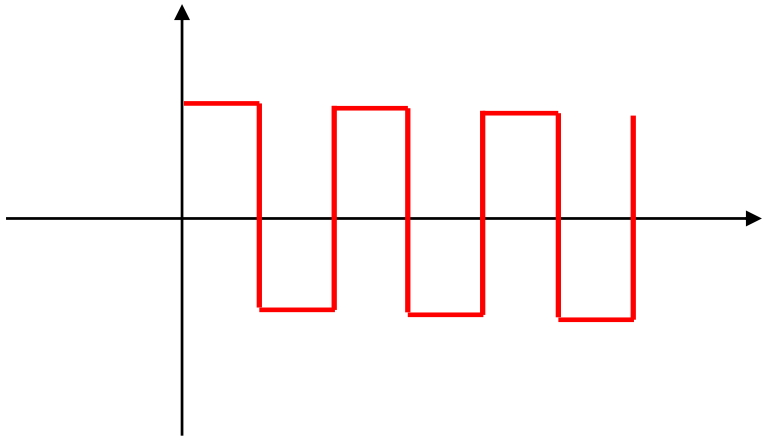
Practical Means to Improve Parameter Estimation Accuracy:

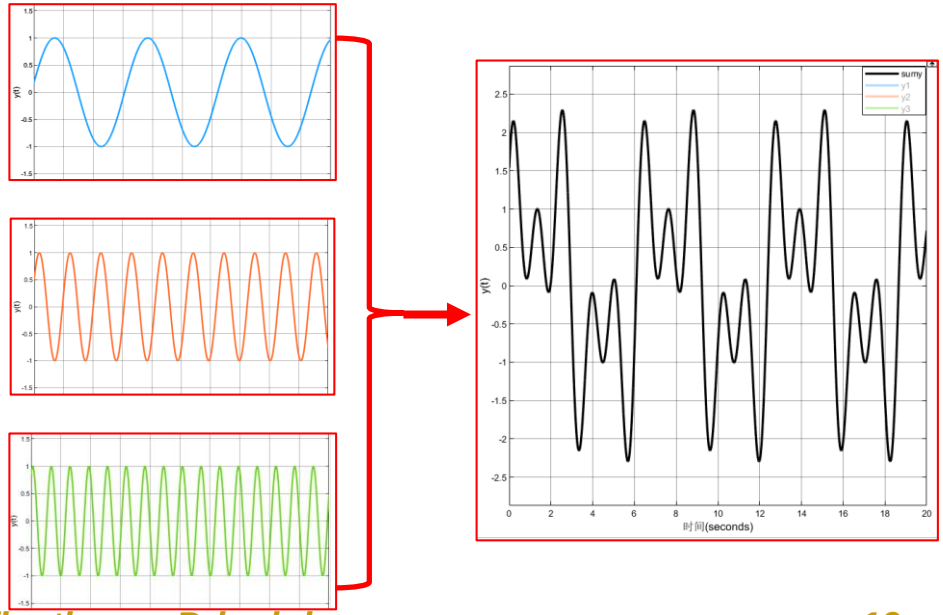
E3. Un-correlate the regressor vector $\varphi(i)$ and the model error vector $e(i)$

System ID of Servo Table Without Rod

■ ID Step 3 — Experimental Design

- ➢ **To identify three model parameters, input signals have to be at least persistently exciting (PE) of order 3.**(为辨识三个模型参数，输入信号必须至少是三阶的持续激励?)
- ➢ **Square wave and sum of sinusoids of several frequencies are commonly used**(通常采用方波和几个不同频率正弦叠加波形)





System ID of Servo Table Without Rod

■ ID Step 3 — Experimental Design (data processing)

- Using as large inputs as possible to increase φ(t) while at the same time making sure that e(t) is kept small（采用尽可能大的输入信号来增大φ(t)）

$$\tilde{\theta}(t) \xrightarrow{t\to\infty} R_\varphi(0)^{-1} R_{\varphi e}(0)$$

inputs ↑ → $R_\varphi(0)$ ↑

SNR ↑ → $R_{\varphi e}(0)$ ↓

→ $\tilde{\theta}(t)$ ↓

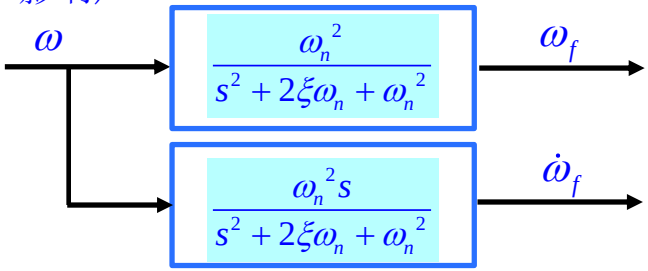
- A low-pass filter identically for each channel is used to reduce measurement noise.（采用对每个通道的同等低通滤波来降低测量噪声影响）

$$\phi(t)^T \vartheta = y$$

Filter at both side

$$\phi(t)_f^{\,T} \vartheta = y_f$$

$$\underbrace{\begin{bmatrix} \dot{\omega} & \omega & \mathrm{sign}(\omega) \end{bmatrix}}_{\phi(t)^T} \underbrace{\begin{bmatrix} \vartheta_1 \\ \vartheta_2 \\ \vartheta_3 \end{bmatrix}}_{\vartheta} = \underbrace{u}_{y}$$



Practical Means to Improve Parameter Estimation Accuracy:

E4 . Maximize the "signal to noise ratio" (SNR)

System ID of Servo Table Without Rod

■ ID Step 4 — Model Validation

- **A good test is to compare the predicted output of the model to measured data that was not used for the fit (called validation data) to study the model's ability to reproduce the behavior of the validation data set by simulation and prediction.** (一个有效的测试方法是将模型预测的输出与未用于拟合的测量数据(称为验证数据)进行对比，以了解模型通过模拟和预测重现验证数据集行为的能力)

Predicted output $\hat{y}(t) = f(u, \hat{\theta})$

Prediction error $e(t) = y(t) - \hat{y}(t)$

- **It is also valuable to look at the data that could not be reproduced by the model (the residuals)** (检查模型无法复现的数据(残差)也很有价值)

The residual $e(t) = y(t) - \varphi(t)^{\mathrm{T}} \hat{\theta}$

Overall System ID Procedure

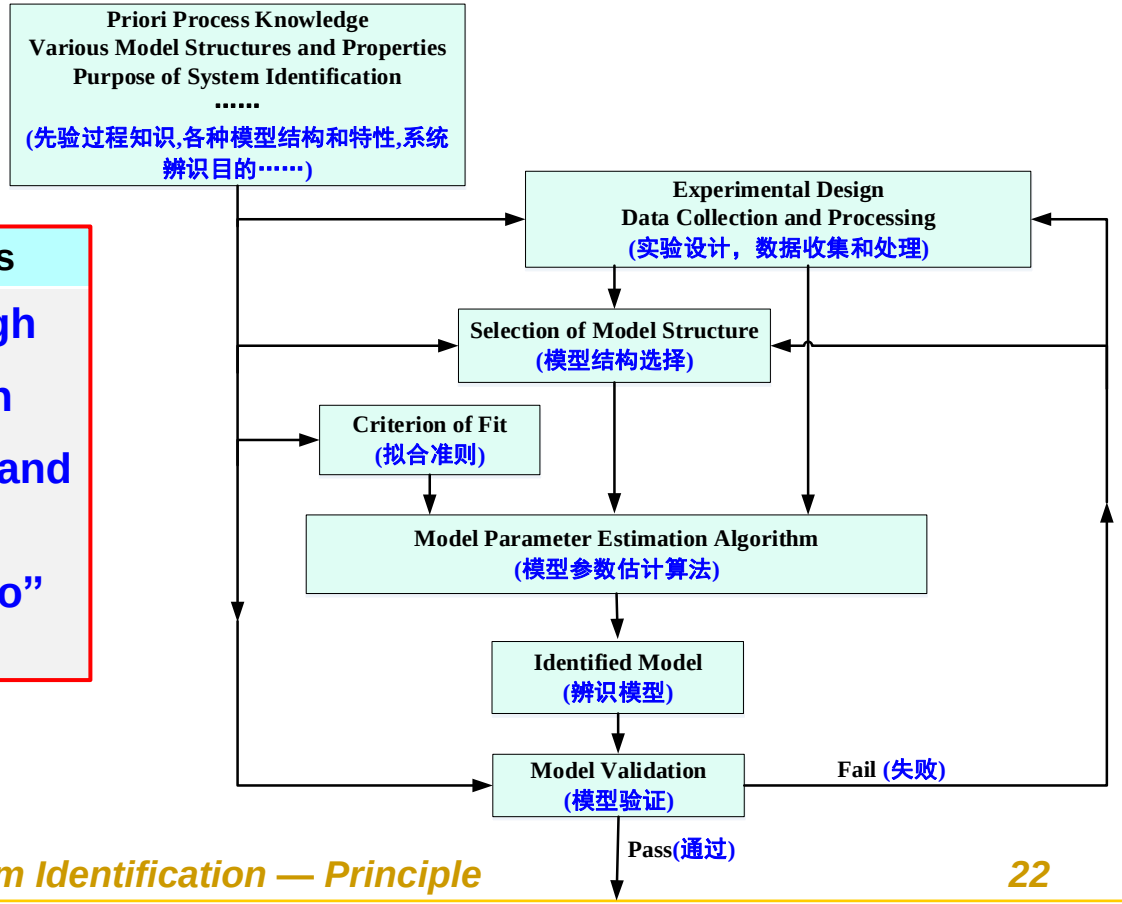
Requirements for good identification results

E1. The data set has to be large enough

E2. The data set has to be rich enough

E3. Un-correlate the regressor vector and the model error vector

E4 . Maximize the "signal to noise ratio" (SNR)



Simulation of System ID for Servo Table

LSE Result

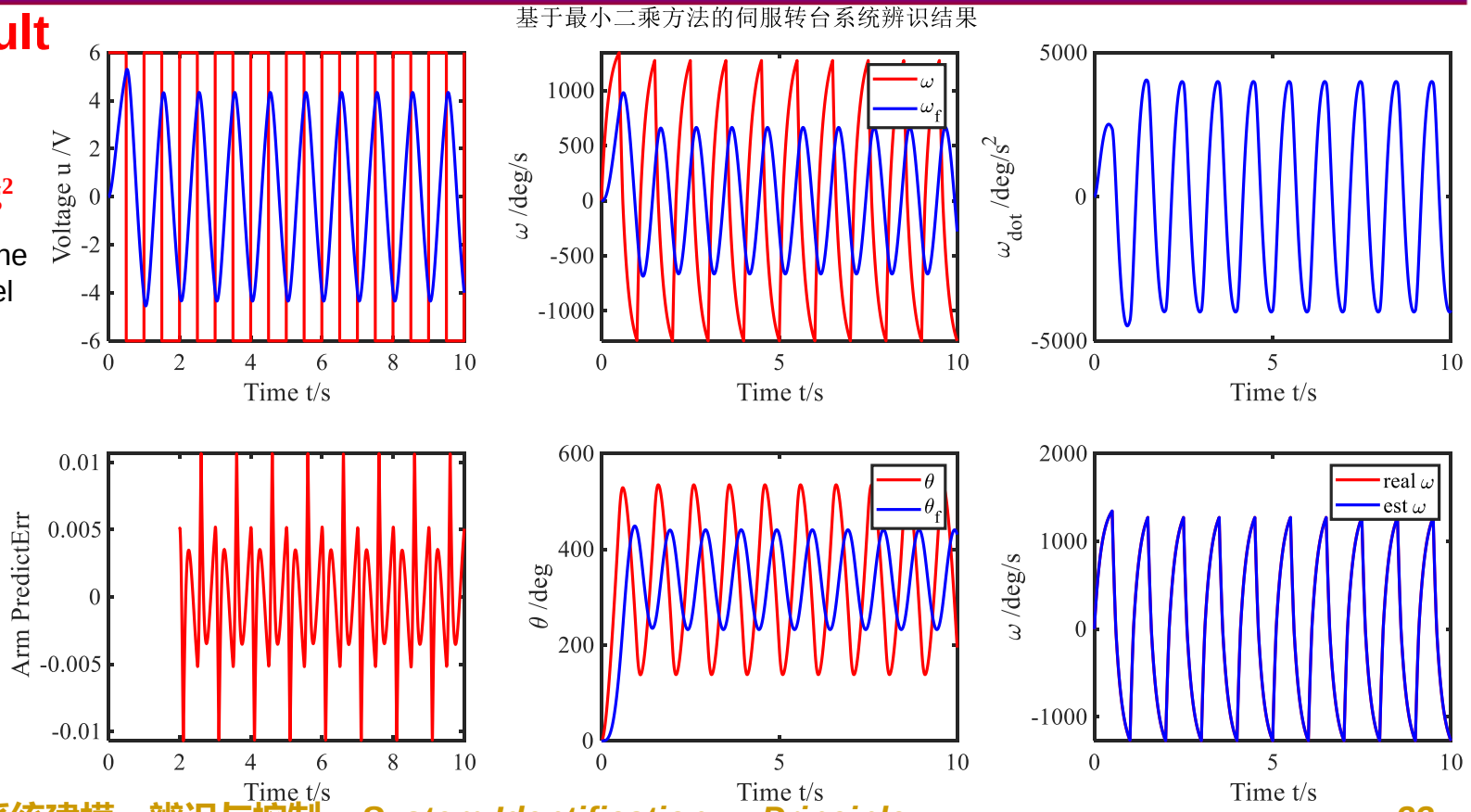
u=6V,σ^2=0deg^2

Parameters of the identified model

$$\bar{K}_t = 4.9754$$

$$\tau_t = 0.1966$$

$$A_c = 0.8753$$



Simulation of System ID for Servo Table

LSE Result

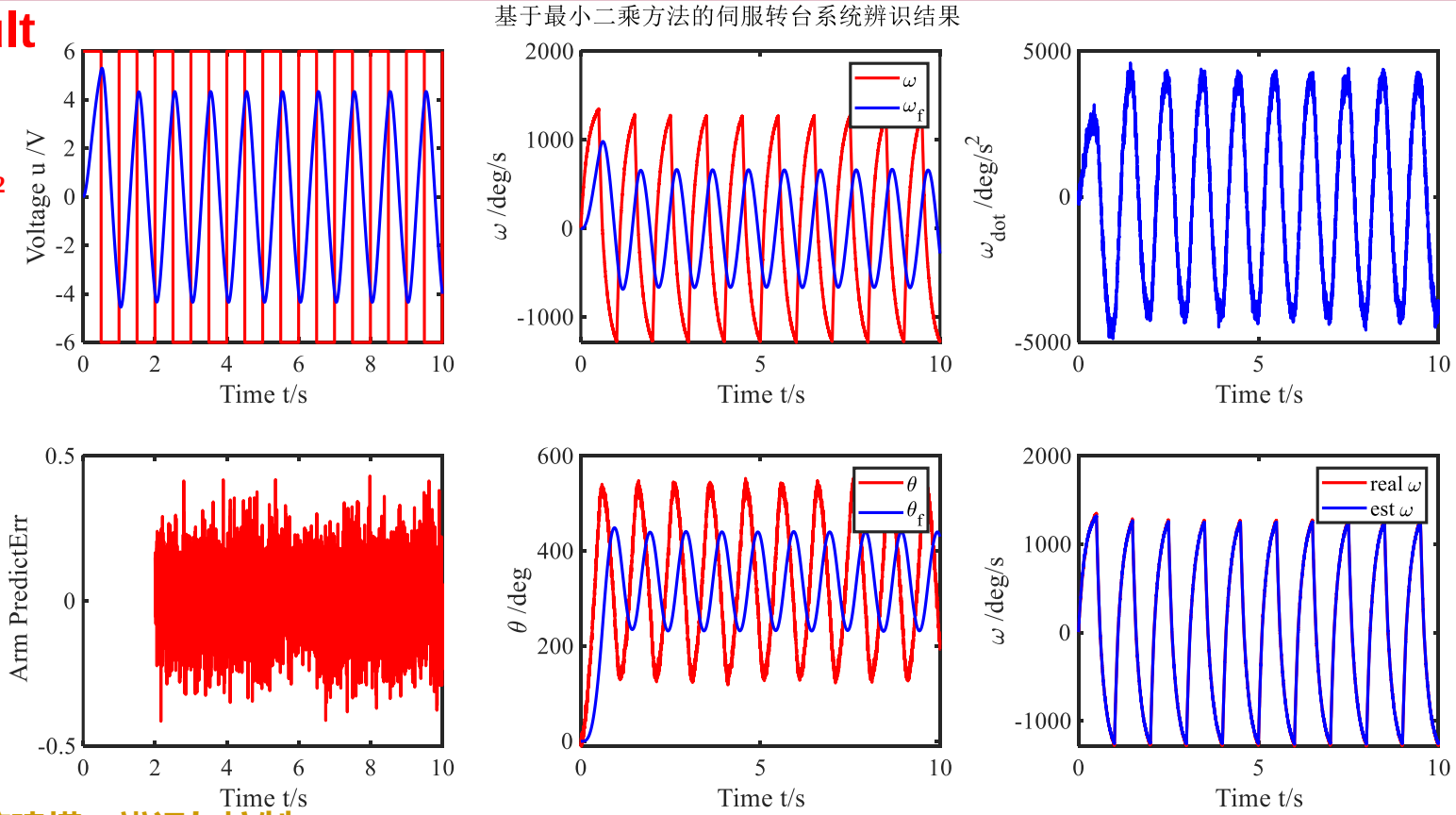
u=6V, σ^2=50deg^2

Parameters of the identified model

$$\bar{K}_t = 4.3821$$

$$\tau_t = 0.1754$$

$$A_c = 0.4464$$



Simulation of System ID for Servo Table

LSE Result

u=1V, σ^2=50deg^2

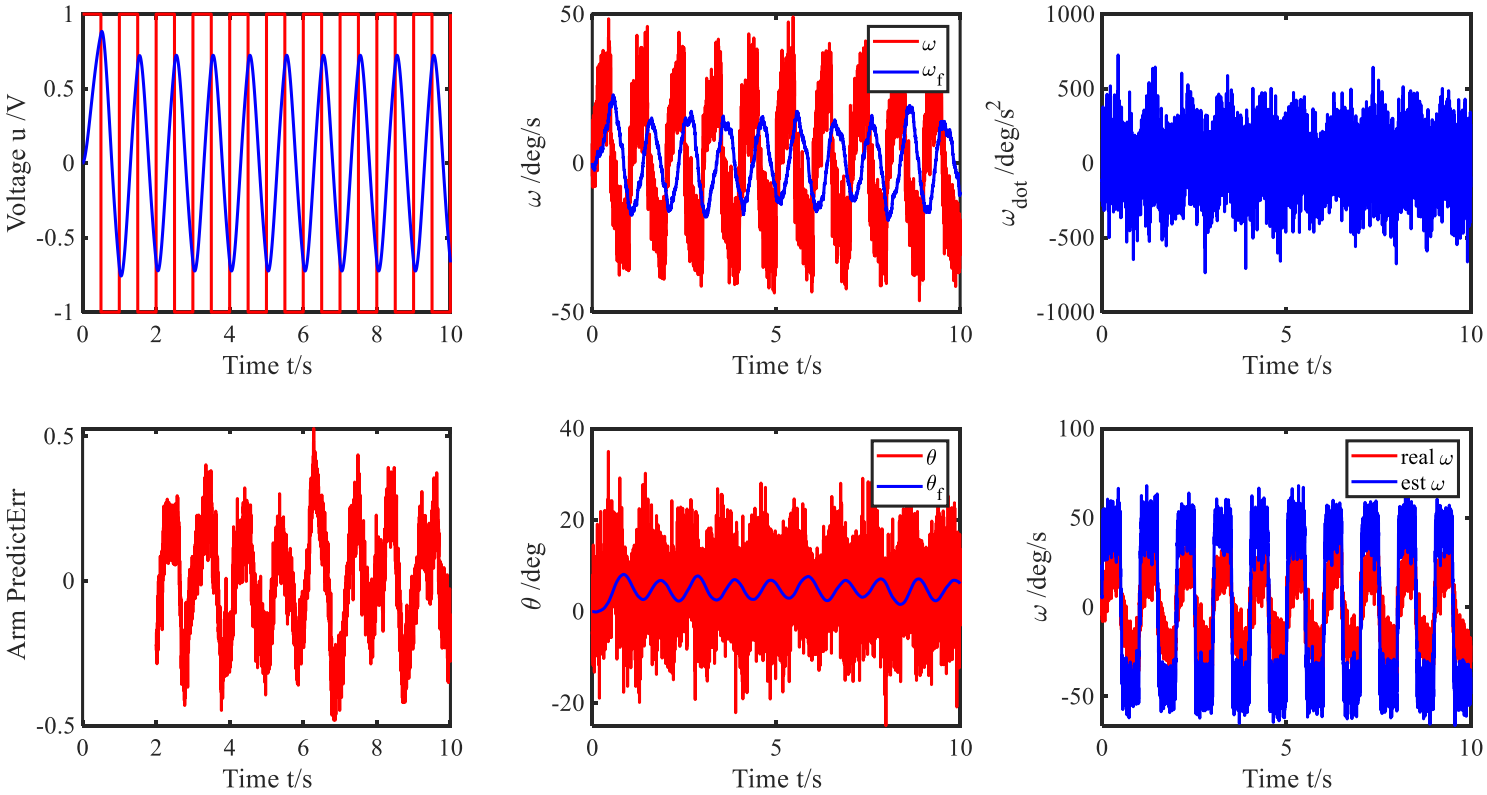
Parameters of the identified model

$$\bar{K}_t = 0.3901$$

$$\tau_t = 0.0077$$

$$A_c = -0.9623$$

基于最小二乘方法的伺服转台系统辨识结果



Simulation of System ID for Servo Table

- System identification of the servo table without rod via LSE with Considering Coulomb Friction Model

Parameter	1V	2V	3V	4V	5V	6V	σ^2	True Value
K_t	2.2634	4.7949	4.9223	4.9546	4.9685	4.9754	0	4.9757
	0.3901	2.0150	2.9452	3.5072	4.0010	4.3821	50	
τ_t	0.2244	0.2036	0.1998	0.1981	0.1972	0.1966	0	0.1938
	0.0077	0.0893	0.1244	0.1444	0.1620	0.1754	50	
A_c	0.7768	0.8498	0.8645	0.8702	0.8735	0.8753	0	0.9
	-0.9623	-0.5894	-0.1554	0.0112	0.2372	0.4464	50	

Dissatisfied! Why?

Experimental System ID of Rotary Servo Table

- **Transfer function from input voltage to the table angle**

$$G_t(s) = \frac{\theta(s)}{u(s)} = \frac{\bar{K}_t}{s(\tau_t s + 1)}$$

I/O model in time domain

$$\tau_t \ddot{\theta}(t) + \dot{\theta}(t) = \bar{K}_t u(s)$$

LSE in time domain

➢ **Regressor model**

$$y = \varphi^T \beta$$

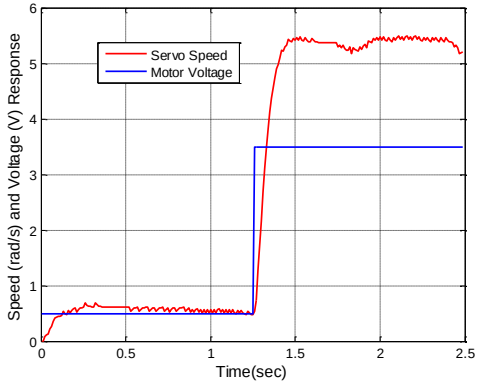
$$y = u$$

$$\varphi = [\ddot{\theta}, \dot{\theta}]^T$$

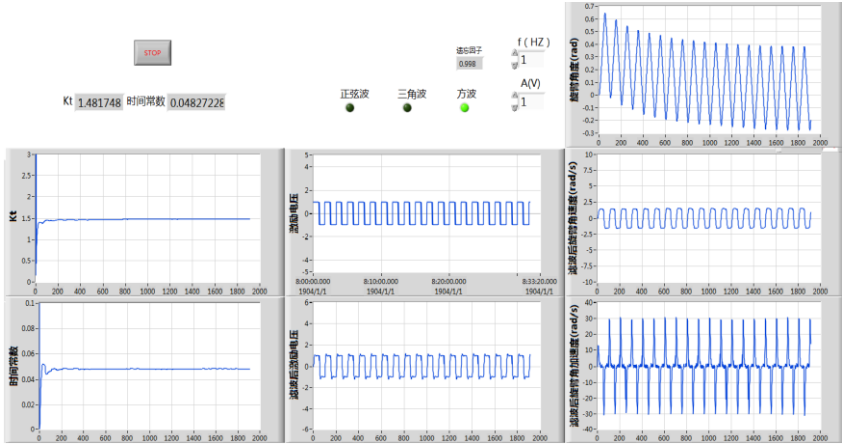
$$\beta = [\tau_t / \bar{K}_t, 1/\bar{K}_t]^T$$

Filter with both sides

$$y_f = \varphi_f^{\,T} \beta$$



Bump Test



Square wave Test

Excitation signals	K_t	τ_t
Nominal values	1.53	0.0252
Sinusoidal	1.38	0.041
Triangle	1.3	0.041
Square	1.48	0.048
Bump Test	1.69	0.066

Physical parameters used in experiments

$J_{eq} = 0.002\text{kg.m}^2$,

$B_{eq} = 0.015\text{N.m.s/rad}$,

$K_g = 70$

System Identification in Frequency-Domain

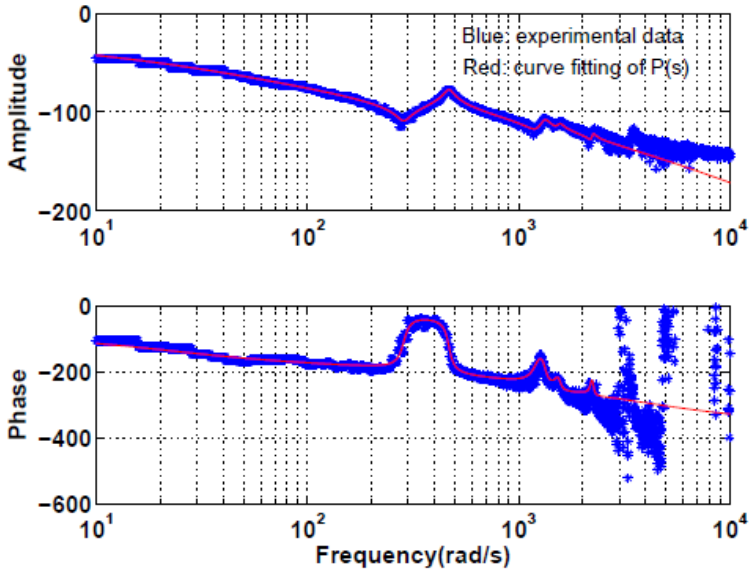
- **Method 3: System Identification in frequency-domain**

  - System ID in time-domain needs a parametrized model structure to start with. The model structure used for the servo table is based on the physics based modeling which assumes rigid-body for all mechanical parts. In reality, all mechanical systems exhibit flexible modes effects at high frequency, so knowing the range of frequency that the previous rigid-body assumption based model is crucial for a successful control design.(时域中的系统辨识需要一个参数化模型结构。上述伺服转台的模型是假设所有机械部件均为刚体物理建模得出的。实际上，所有机械系统在高频时均会表现出柔性模态效应，因此了解基于刚体假设的模型所适用的频率范围对于成功实现控制设计至关重要)

System Identification in Frequency-Domain

- **Method 3: System Identification in frequency-domain**

➢ System ID in frequency-domain introduced below only assumes the underlying dynamics can be adequately described by an LTI system, which includes any flexible modes effect.（下面介绍的频域系统辨识仅假设可由线性时不变(LTI)系统来充分描述系统动态，包括任何柔性模态影响）



Experimental Design

- **Identifiability Requirement**
  - Input signals used for experimental tests need to be rich enough to cover entire frequency-range of interests, which can be done by（用于实验测试的输入信号要足够丰富，以便覆盖整个感兴趣的频率范围，这可以通过以下方式实现）
    - Multiple tests with sinusoidal inputs of single frequency spanning the range, or（单频正弦输入范围内的多次测试）
    - Single test with a rich complicated input (e.g., periodic input of multi-sinusoids)（具有丰富复杂输入的单一测试(例如，多正弦曲线的周期性输入)）
- **Minimizing Model Error Effect**
  - Choose input signals large enough to have good signal-to-noise ratio and to reduce the effect of unavoidable nonlinear elements in experiments (e.g., Coulomb friction)(选择足够大的输入信号，以具有良好的信噪比，并减少实验中不可避免的非线性因素（如库仑摩擦）的影响)
  - While not being overly large that they induce additional model error by violating the underline linear assumptions (e.g., input and state saturations due to too large input signals)(但不能太大以免违反线性假设(例如，由于输入信号太大而导致的输入和状态饱和)，从而引起额外的模型误差)

Experiments and Data Processing

- **Steady-State Response Requirement**

  Conduct experiments with the test input signals long enough to make sure that the recorded measurement responses do converge to their steady-state responses!（使用足够长的测试输入信号进行实验，以确保记录的测量响应收敛到其稳态响应）

- **Experimental Tests with Sinusoids of Single Frequency**

  At each frequency ω_i of the sinusoidal inputs, obtain the magnitude ratio M_i and phase shaft ϕ_i of the steady-state sinusoidal response $y_{ss} = A_y \sin(\omega_i t + \phi_i)$ to the input $u = A_u \sin(\omega_i t)$ to get $G(j\omega_i)$（每次单一频率正弦信号测试）

$$\begin{aligned} |G(j\omega_i)| &= M_i \\ \angle G(j\omega_i) &= \phi_i \end{aligned} \Rightarrow \begin{cases} \|G(j\omega_i)\| = 20\log_{10}(M_i) \\ \angle G(j\omega_i) = \phi_i \times \dfrac{180^o}{\pi} \end{cases}$$

where

$$M_i = \frac{A_y}{A_u}, \phi_i = \omega_i \Delta t$$

Frequency Response via System ID Tool Box

- **Experimental Test**
  - Does not need to perform many sinusoidal tests at different frequencies. Instead, generate a periodic input $u(t)$ that contains components of all interested frequencies. Apply this periodic input and obtain the *steady-state* periodic response $y(t)$ via one experimental test（不需要在不同频率下进行多次正弦测试。生成一个包含所有感兴趣频率分量的周期性输入$u(t)$ 。应用该周期性输入并通过一次实验测试获得稳态周期响应$y(t)$）

- **Data Processing**
  - Obtain the Fourier Transform of input, $U(j\omega)$, and output, $Y(j\omega)$, respectively
  - Obtain Frequency Response by

$$G(j\omega) = \frac{Y(j\omega)}{U(j\omega)}$$

MATLAB System ID Tool Box

- **Input Signal Generation**

Freq. Resolution : $\frac{1}{N} f_s$ Hz

Data Length : $N \frac{1}{f_s}$ sec

>> help idinput

IDINPUT Generates input signals for identification

U = IDINPUT(N,TYPE,BAND,LEVELS)

U: The generated input signal. A column vector or a N-by-nu matrix

N: The length of the input

N = [N Nu] gives a N-by-Nu input (Nu input channels)

N = [P Nu M] gives a M*P-by-Nu input, periodic with period P and with M **periods**

Default values are Nu = 1 and M =1;

TYPE: One of the following:

'RGS': Generates a Random, Gaussian Signal

'RBS': Generates a Random, Binary Signal

'PRBS': Generates a Pseudo-random, Binary Signal

'SINE': Generates a sum-of-sinusoid signal

Default: TYPE = 'RBS'

MATLAB System ID Tool Box

BAND: A 1 by 2 row vector that defines the frequency band for the input‘s frequency contents

For the 'RS', 'RBS' and 'SINE' cases BAND = [LFR,HFR], where LFR and HFR are the lower and upper limits of the passband, expressed in fractions of the Nyquist frequency (thus always numbers between 0 and 1).

For the 'PRBS' case BAND = [0,B], where B is such that the signal is constant over intervals of length 1/B (the Clock Period).

Default: BAND =[0 1].

LEVELS = [MI, MA]: A 1 by 2 row vector, defining the input levels

For 'RBS', 'PRBS', and 'SINE', the levels are adjusted so that the input signal always is between MI and MA.

For the 'RGS' case, MI is the signal's mean value minus one standard deviation and MA is the signal's mean plus one standard deviation.

Default LEVELS = [-1 1].

In the 'PRBS' case, if M > 1, the length of the data sequence and the period is adjusted so that always an integer number of maximum length PRBS periods are obtained. If M = 1 the period is chosen so that it becomes longer than P = N. In the multiinput case the signals are maximally shifted. This means that P/Nu is an upper bound for the model orders that can be used to identify systems excited by such a signal.

MATLAB System ID Tool Box

In the 'SINE' case, the sinusoids are chosen from the frequency grid freq = 2*pi*[1:Grid_Skip:fix(P/2)]/P intersected with pi*[BAND(1) BAND(2)]. For multi-input signals, the different inputs use different frequencies from this grid. An integer number of full periods is always delivered. The selected frequencies are obtained as [U,FREQS] = IDINPUT(....), where row ku of FREQS contains the frequencies of input number ku.

The resulting signal is affected by a 5th input argument SINEDATA

U = IDINPUT(N,TYPE,BAND,LEVELS,SINEDATA)

where

SINEDATA= [No_of_Sinusoids, No_of_Trials, Grid_Skip],

$$\sum_{i=1}^{N} \sin(\omega_i t + \phi_i)$$

meaning that No_of_Sinusoids are equally spread over the indicated BAND, trying No_of_Trials different, random, relative phases, until the lowest amplitude signal is found

Default SINEDATA = [10,10,1];

Generation of Exciting Signals for System ID

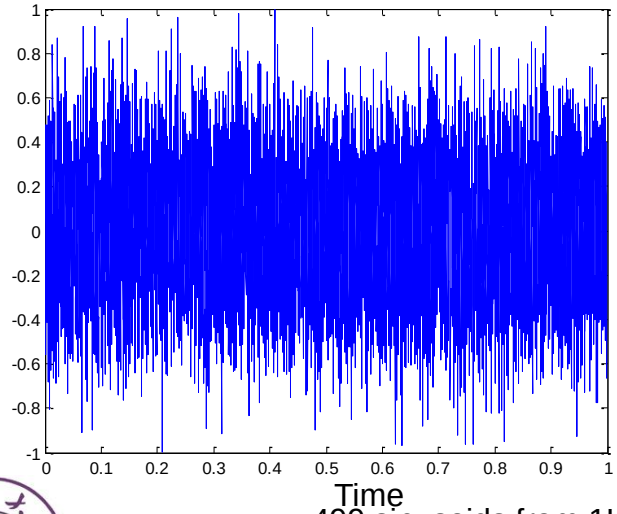
"***idinput***" command to generate a sum of 1000 sinusoids ranging from 1Hz to 5KHz with a sampling frequency of 10KHz :

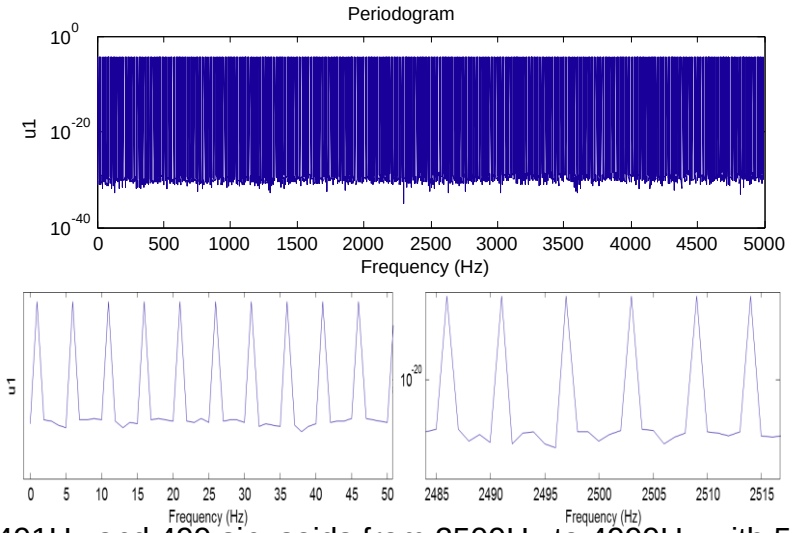
sinesum = idinput([10000 1 1],'sine', [0 1],[-1 1],[1000 50 1]);

Freq. Resolution : $\frac{1}{N} f_s$ Hz

Data Length : $N \frac{1}{f_s}$ sec

Sampling frequency : f_s=10KHz

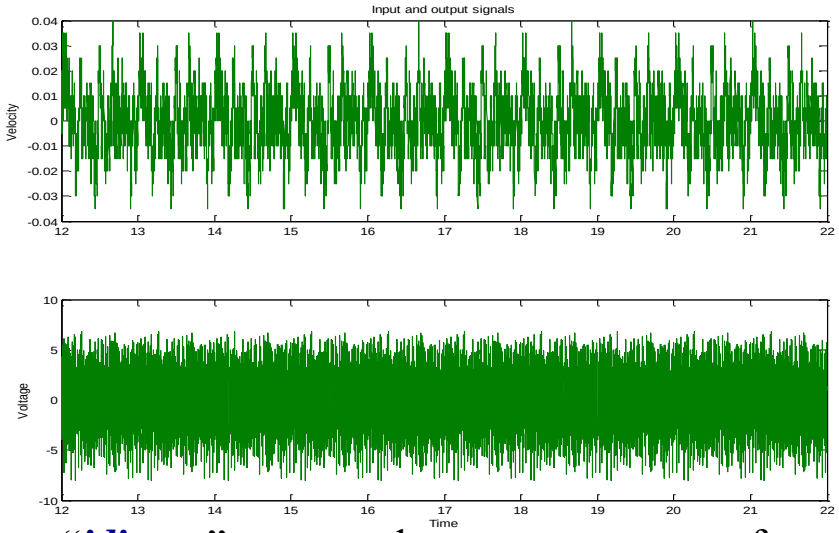




499 sinusoids from 1Hz to 2491Hz and 499 sinusoids from 2509Hz to 4999Hz with 5Hz spacing, 2 sinusoids at 2497Hz and 2503Hz with 6Hz spacing

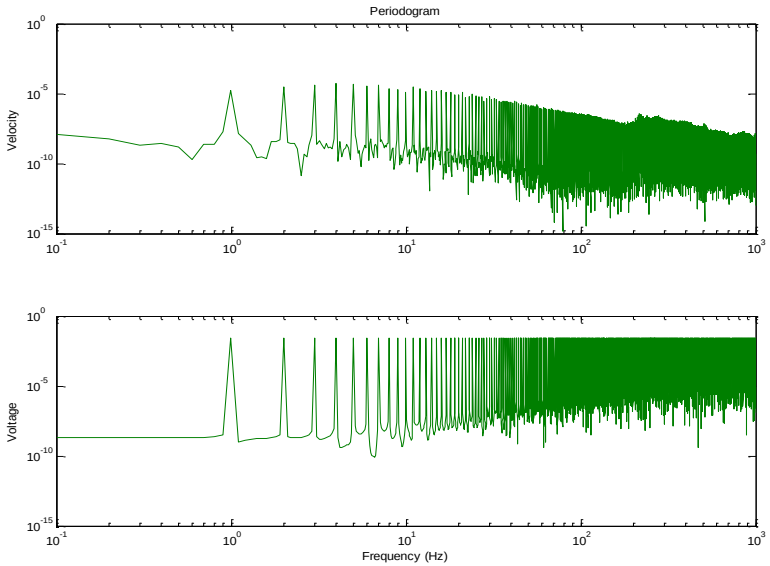
System Dynamics Identification

- **Single Sinusoidal Test**



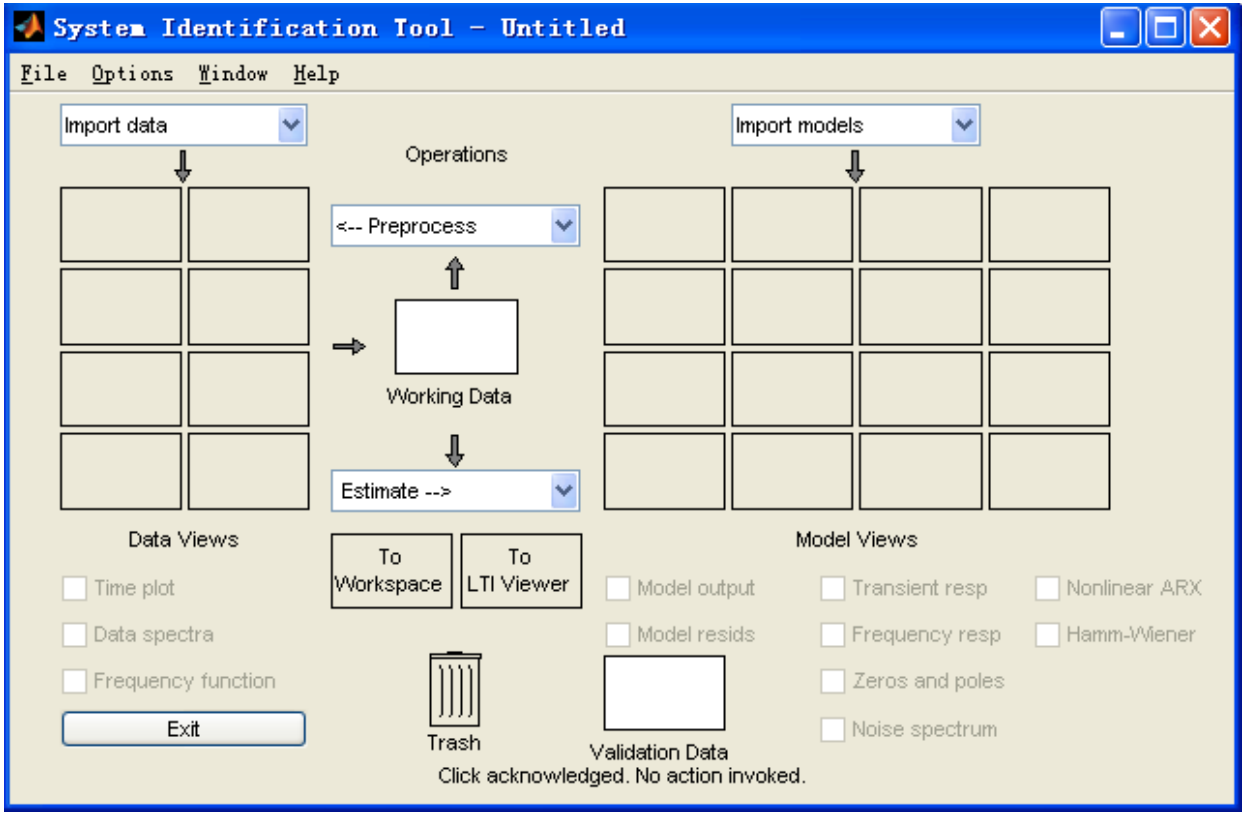
"***idinput***" command to generate a sum of 1000 sinusoids ranging from 1Hz to 1KHz with a sampling frequency of 10KHz :

sinesum = idinput([10000 1 1],'sine', [0 0.2], [-1 1],[1000 50 1])



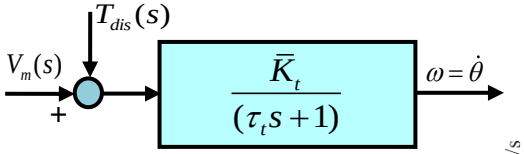
Spectrum of Input/Output using MATLAB "***fft***"

System Identification Tool Box GUI



ident

Simulation Frequency Identification of Servo Table



Servo table dynamics

Magnitude of input signal = 10V,
frequency range = [0.01,30]Hz,
Number of sinusoids:225

Estimated Value

$\bar{K}_t = 1.8647$

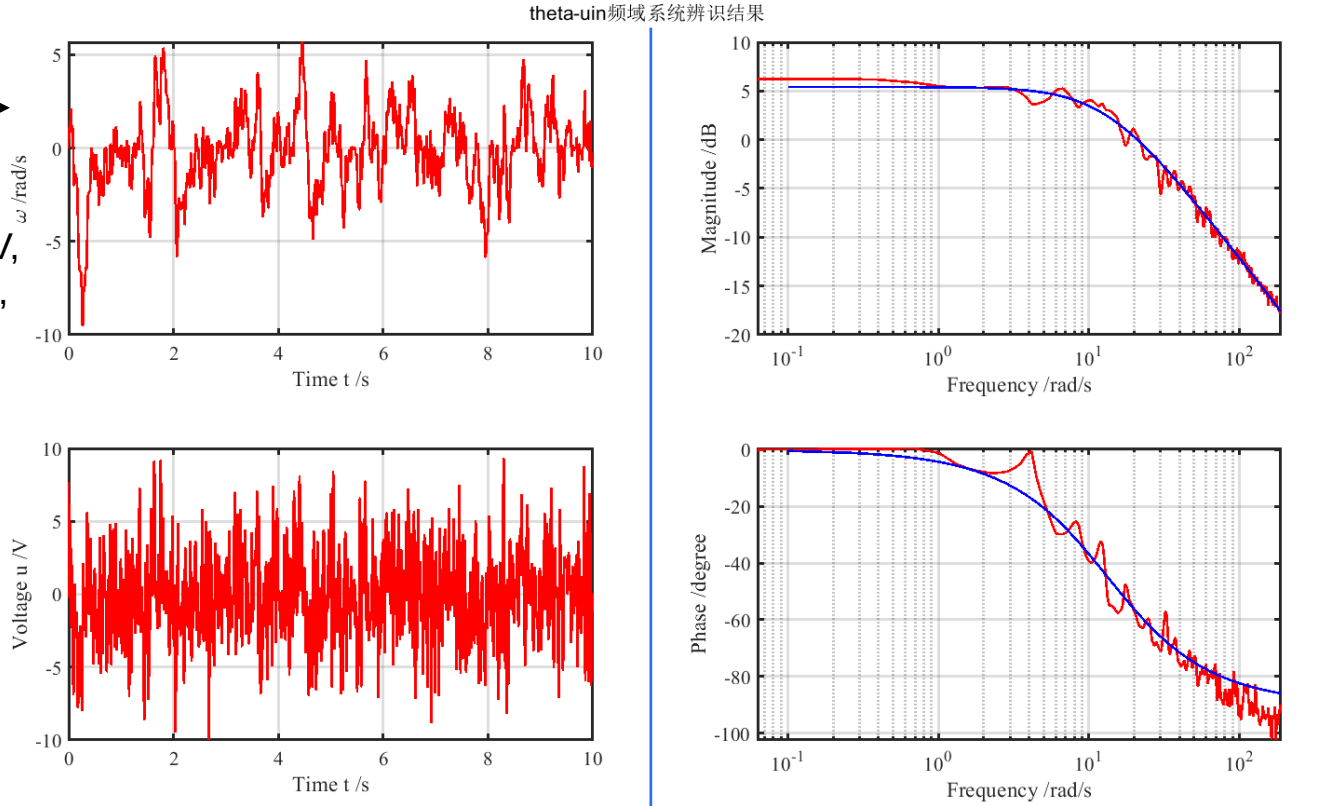
$\tau_t = 0.0749$

True Value

$\bar{K}_{t0} = 4.9757$

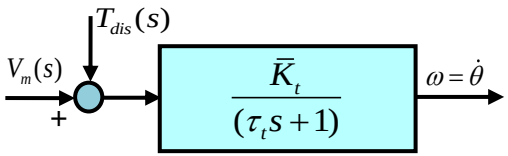
$\tau_{t0} = 0.1938$

$T_{dis0} = 0.9\,\mathrm{sgn}(\omega)$



Input and Output Signals (Time Domain) | **Input and Output Signals (Periodogram)**

Simulation Frequency Identification of Servo Table



Servo table dynamics

Magnitude of input signal = 10V, frequency range = [0.01,2]Hz, Number of sinusoids:15

Estimated Value

$\bar{K}_t = 3.9843$

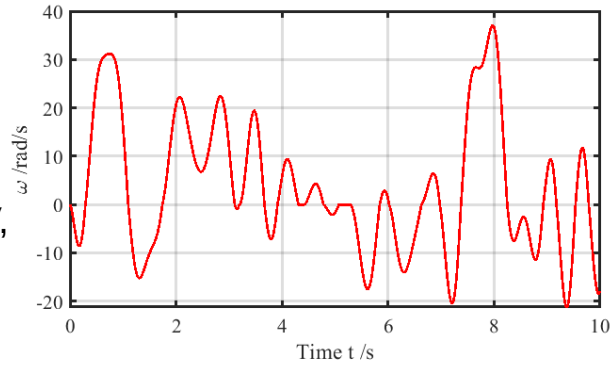
$\tau_t = 0.1582$

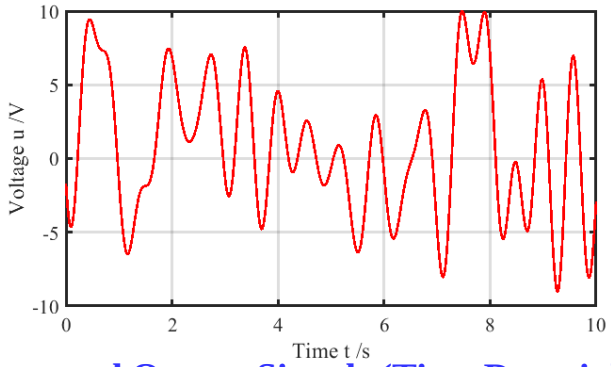
True Value

$\bar{K}_{t0} = 4.9757$

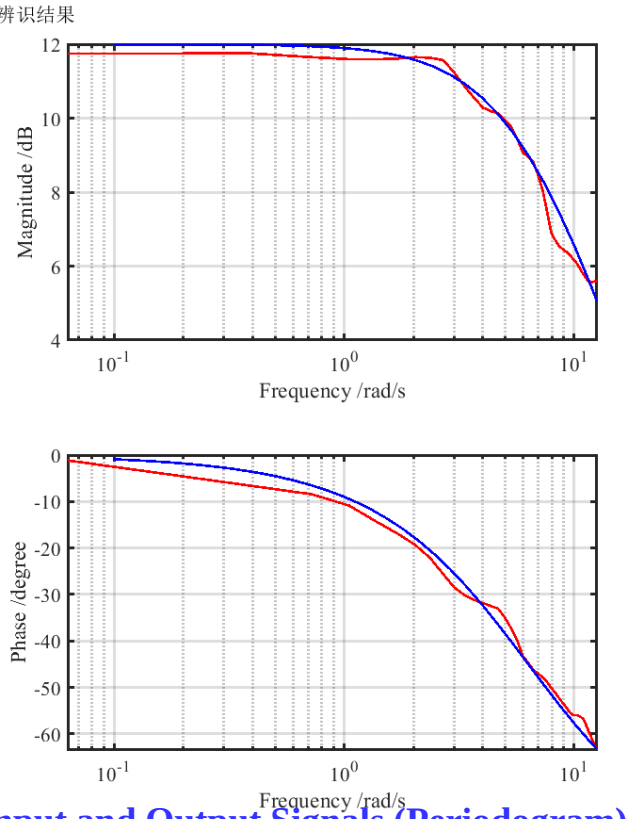
$\tau_{t0} = 0.1938$

$T_{dis0} = 0.9\,\mathrm{sgn}(\omega)$





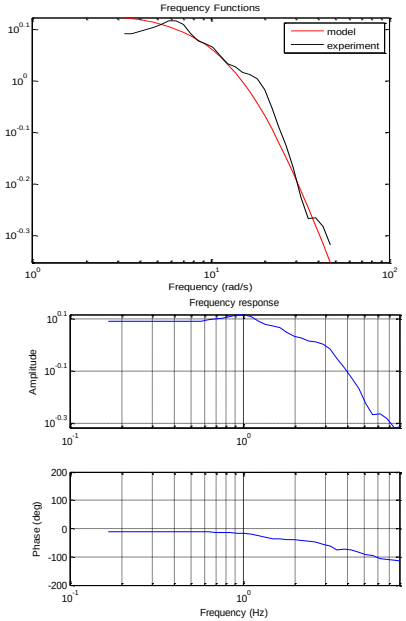
Input and Output Signals (Time Domain)



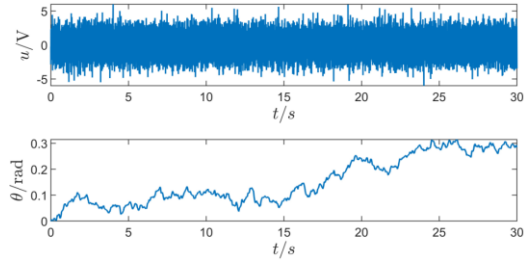
Input and Output Signals (Periodogram)

Experimental Frequency Identification of Rotary Servo Table

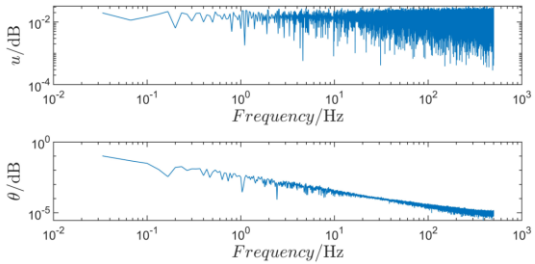
$$G_\omega(s) = \frac{\Omega(s)}{u(s)} = \frac{K_r}{\tau_r s + 1}$$



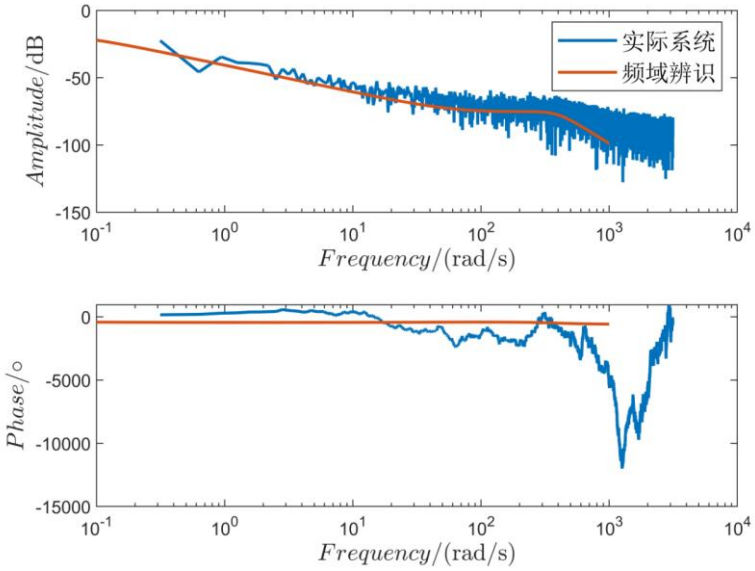
The excitation signal is a chirp signal with frequency ranging from 0.005Hz − 8Hz



Signals in time domain



Signals in frequency domain



Frequency identification

The excitation signal is a sum of sinusoids with chirp signal with frequency ranging from 0.5Hz − 500Hz and amplitude =5V

Parameters	K_r	τ_r
Nominal values	1.53	0.0252
Frequency response(sin)	1.53	0.0571
Frequency response(sum of sin)	1.27	0.059
Bump Test	1.69	0.066